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# Exam PL-300

Microsoft Power BI  
Data Analyst Associate  
Crash Course

Day 1

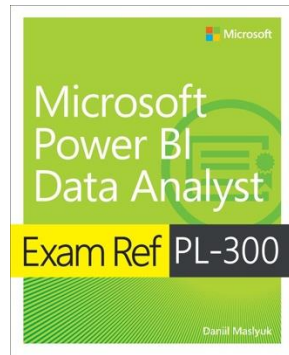
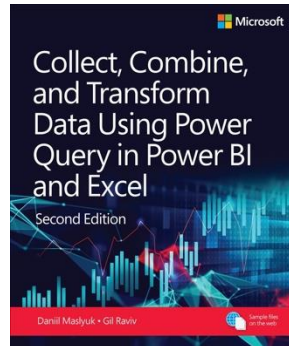
Presented by  
Daniil Maslyuk



# Daniil Maslyuk

Your presenter today

- Independent consultant
- Author of all the official Power BI exam reference books
- Author of *Collect, Combine, and Transform Data Using Power Query in Power BI and Excel, 2<sup>nd</sup> Edition*
- Technical Editor of *The Definitive Guide to DAX*
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# Your Power BI experience

How long have you been using Power BI for?

- Less than 1 month
- $1 \text{ month} \leq X < 3 \text{ months}$
- $3 \text{ months} \leq X < 1 \text{ year}$
- $1 \text{ year} \leq X < 2 \text{ years}$
- $2 \text{ years} \leq X < 5 \text{ years}$
- 5 years or more

# Course overview

- Day 1
  - Segment 1: Prepare the data
  - Segment 2: Model the data
- Day 2
  - Segment 3: Model the data, continued
  - Segment 4: Create reports
  - Segment 5: Manage and secure Power BI

# Exam format

- Proctored exam
- 100 minutes
- Different question types
  - Case studies
  - Multiple choice
  - Drag and drop
  - Build list
- Free renewals!

# Prepare the data

## Segment 1

- Get or connect to data
- Profile and clean the data
- Transform and load the data



# Get or connect to data

## Segment 1: Prepare the data

- Identify and connect to data sources or a shared semantic model
- Change data source settings, including credentials and privacy levels
- Choose between DirectQuery and Import
- Create and modify parameters



# Profile and clean the data

Segment 1: Prepare the data

- Evaluate data, including data statistics and column properties
- Resolve inconsistencies, unexpected or null values, and data quality issues
- Resolve data import errors





# Transform and load the data

## Segment 1: Prepare the data

- Select appropriate column data types
- Create and transform columns
- Group and aggregate rows
- Pivot, unpivot, and transpose data
- Convert semi-structured data to a table
- Create fact tables and dimension tables
- Identify when to use reference or duplicate queries and the resulting impact
- Merge and append queries
- Identify and create appropriate keys for relationships
- Configure data loading for queries

# Knowledge check: Question 1

Segment 1: Prepare the data

You are merging Product and Product Category in Power Query. Some products do not have any corresponding product category.

You need to ensure that all Product rows are shown.

Which join type should you use?

- A. Left outer
- B. Inner
- C. Left anti
- D. Full outer

# Knowledge check: Question 2

Segment 1: Prepare the data

You are previewing sales data from a flat file in Power Query. The file has years on rows, months on columns, and units as values.

You need to shape the data into the right format for Power BI.

What should you do?

- A. Promote headers
- B. Transpose
- C. Pivot
- D. Unpivot

# Knowledge check: Question 3

Segment 1: Prepare the data

You are importing data from an Excel sheet, where the data with column headers starts on row 3. The first column also contains some null values.

You need to remove the rows before column headers and use row 3 as column headers.

What should you do first?

- A. Filter out empty values from the first column
- B. Demote headers
- C. Remove the first two rows
- D. Promote headers

# Model the data

## Segment 2

- Design and implement a data model
- Create model calculations by using DAX
- *Optimize model performance (covered tomorrow)*



# Design and implement a data model

## Segment 2: Model the data

- Configure table and column properties
- Implement role-playing dimensions
- Define a relationship's cardinality and cross-filter direction
- Create a common date table
- Identify use cases for calculated columns and calculated tables



# Create model calculations by using DAX

## Segment 2: Model the data

- Create single aggregation measures
- Use the CALCULATE function
- Implement time intelligence measures
- Use basic statistical functions
- Create semi-additive measures
- Create a measure by using quick measures
- Create calculated tables or columns
- Create calculation groups



# Optimize model performance

Segment 2: Model the data

- Covered tomorrow 😊
- Knowledge check is also tomorrow!



# Wrap-up of Day 1

## Segment 3

Covered today:

- Segment 1: Prepare the data
- Segment 2: Model the data

Covered tomorrow:

- Segment 3: Model the data, continued
- Segment 4: Create reports
- Segment 5: Manage and secure Power BI
- Final tips for the exam



# See you tomorrow!

Please provide your course feedback

